



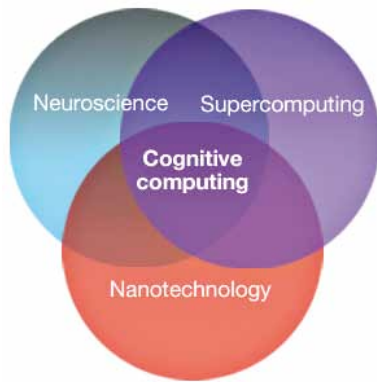
Wikiweb for iOS

It is more than just a Wikipedia reader, visualizing connections between articles with awesome presentation



Webcam Toy

Want to have some fun with your webcam? Webcam Toy allows you to use it with some cool effects



The IBM SyNAPSE Initiative: Blurring the lines between neural and digital

breakingly new on these fronts as of now, but the mere fact that these areas of study exist, point to the fact that there's something out there. And that we will get to it sooner or later.

The Direct Comparison

Non-volatile memory (based on human memory), processors with in-built memory (the brain), then comes DNA computation, and now we know that even DNA can store massive amounts of data. Maybe nature built us that way - biological supercomputers?

The brain:

The processor (duh). And also the RAM, storage unit and any other interfacing chip that you may or may not know of. Seriously, it is an amazing organ that the world's best computers can only hope to emulate and maybe, just maybe, hope to match in the years to come (they wish!). A fully developed human brain can store 2.5 petabytes of data. That should amount to approximately a million gigabytes. You can gain some perspective from the fact that just to use up all that space, you would have to store over 300 years of conscious data (and still we carry chits to exams, tch-tch).

Now for the hard talk - computing power. By some estimates it takes only one human brain, operating at full capacity, to compete with all of the world's supercomputers combined. This does not apply to specific task-oriented machines such as, say, those designed to play chess though. So much for the quantitative analysis, it is the qualitative one that

crowns the brain as the ultimate champion of signal processing. That walnut shaped blob of neurons really is a masterpiece of biological evolution. One that might take us hundreds, if not thousands, of years to replicate.

To get some perspective here, we can see that game developers have employed a program that was designed to assist spaceships for landing on the moon, just to simulate the smooth movements of a human being. Then there's breathing, body fluid control and so much else, just in the involuntary part. This still leaves us with enough processing power for us to get through our day. Then there's also the cherry on the cake - we are capable of learning. Cognitive abilities are inherent in humans, and ancient records of the forbidden experiment, raising a child without the influence of culture, show that even a human with little to no 'herd' contact will develop some sort of a surrounding-based intelligence. Something a computer has never been able to do.

Other Storage Centres:

If we really are supercomputers, we are definitely the best, that is a given. The reason is that we have overcome the one difficulty that our digital counterparts still have to reach - we are capable of reproduction and thus ensure that our race lives on. For this purpose, we store lots of 'unchangeable' data inside our bodies - think of it as 'ROMs' in our body, as has been made popular by online memes (and indeed one of the few 'truths' that they propagate). One study claims that a regular chromosome in a human body carries about 75 MB of data, and about half of that is the capacity of a sperm cell - about 37.5 MB. So, in order to create a new life (read: computer), the entire 'initialization factor set' is compressed and passed on when a child is



Now this is really, really compact storage

conceived. It is from the decompression and breakdown of that information, and usage of complementary data in the female body (at least another 37.5MB), that a child is 'prepared'. Freakishly outrageous, but freakishly correct. True story.

Nerves:

Connecting wires to our peripherals. Nothing fancy here. In fact, the speed of transmission of the signal isn't even in comparison with the digital counterparts. 120 meters per second, there's quite a bit of lag there.

The Final Word

The popular base on which humans claim superiority over machines is their 'intelligence', or decision making ability, out of their own free will. Now as a treat to you all before we go, we present this startling fact - scientifically, there is nothing called free will. Yes, we indeed are, without a shred of doubt, computed and computing systems. The entire discussion that went down here was to draw parallels with computing as we know it, but the fact of the matter is that computing does not have to be the way that we know it. Any system that is made up of components that can hold distinct values, and can respond to 'inputs' and produce 'outputs' is theoretically a computing system.

Physics says that if we could get to know the exact state of any and all particles in a system, then the exact state of the system can be predicted, and that this statement holds at all scales of existence. Our world is made up of, at its most basic level of atoms. Stephen Hawking, in the legendary book *A Brief History of Time* claims that if we could, somehow get to know the states of all the particles in the Universe, then we could predict everything at any point of time in the future, and calculate everything at point in the past. Computers, to us, seem unintelligent because we know all of the states, and all of the inputs of a program. Well, if the programmer we talked about last time knows all the states and all the inputs, we are no better than our idiot boxes to him. There goes that human pride down the drain. Ouch! 